N-gram Activity Sheet

Three short problems on backoff, deleted interpolation, and entropy

Corpus for all activities (19 tokens): the cat sat on the mat . the dog sat on the rug . the cat ate fish .

Activity 1: Backoff (3 marks)

For each phrase, state which level (trigram, bigram, or unigram) you used. Show your working.

a) "the cat sat" (1 mark)

Answer: Trigram

Working:

Count("the cat sat") = 1; Count("the cat") = 2.

P(sat | the cat) = Count(the cat sat) / Count(the cat) = 1/2 = 0.5.

Level used: Trigram.

b) "dog ate fish" (1 mark)

Answer: Unigram (backoff)

Working:

The trigram "dog ate fish" does not occur, and the bigram "dog ate" does not occur. Back off to the unigram level.

Count(fish) = 1 out of 19 tokens, so P(fish) = 1/19 ≈ 0.0526.

Level used: Unigram.

c) "dog sat fish" (1 mark)

Answer: Unigram (backoff)

Working:

The trigram "dog sat fish" does not occur. The bigram "dog sat" occurs once, but the next word after "dog sat" is "on," not "fish." Back off to the unigram level for "fish."

P(fish) = 1/19 ≈ 0.0526.

Level used: Unigram.

Activity 2: Deleted Interpolation (6 marks)

Given: $\lambda_1 + \lambda_2 + \lambda_3 = 1$

Part A (2 marks)

Using $\lambda_1 = 0.1$, $\lambda_2 = 0.3$, $\lambda_3 = 0.6$, find P(sat | the cat).

Interpolation formula:

$P = \lambda_1 P(\text{unigram}) + \lambda_2 P(\text{bigram}) + \lambda_3 P(\text{trigram})$

From the corpus: P(sat) = 2/19; P(sat | cat) = 1/2; P(sat | the cat) = 1/2.

P = 0.1(2/19) + 0.3(1/2) + 0.6(1/2)

P = 0.01053 + 0.15 + 0.30 = 0.46053.

Answer: P(sat | the cat) ≈ 0.46053.

Part B (4 marks)

Held-out sentence: "the dog ate fish" (trigrams: "the dog ate", "dog ate fish"). Compare Set 1 λ = (0.1, 0.3, 0.6) and Set 2 λ = (0.3, 0.6, 0.1). Find each trigram's interpolated probability and add the natural logs (ln). Which set would deleted interpolation choose, and why?

Assumption for this calculation: use the maximum-likelihood n-gram probabilities from the given corpus; unseen bigram/trigram probabilities are zero. The unigram probability of "ate" and "fish" is 1/19 each.

Trigram 1: "the dog ate"

P(ate) = 1/19; P(ate | dog) = 0; P(ate | the dog) = 0.

Set 1: P_1 = 0.1(1/19) + 0.3(0) + 0.6(0) = 0.005263.

Set 2: P_2 = 0.3(1/19) + 0.6(0) + 0.1(0) = 0.015789.

Trigram 2: "dog ate fish"

P(fish) = 1/19; P(fish | ate) = 0; P(fish | dog ate) = 0.

Set 1: P_1 = 0.1(1/19) + 0.3(0) + 0.6(0) = 0.005263.

Set 2: P_2 = 0.3(1/19) + 0.6(0) + 0.1(0) = 0.015789.

Sum of natural logarithms

Set 1: ln(0.005263) + ln(0.005263) ≈ −10.491.

Set 2: ln(0.015789) + ln(0.015789) ≈ −8.301.

Conclusion: Set 2 is preferred because its total log probability is higher (less negative), so it assigns a higher probability to the held-out sentence.

Activity 3: Entropy and Perplexity (6 marks)

Entropy formula: $H = -\Sigma\, p \log_2 p$. Perplexity: $PP = 2^H$.

Part A (2 marks)

Weather: Sunny p = 0.5, Cloudy p = 0.25, Rainy p = 0.25.

$H = -[0.5 \log_2 (0.5) + 0.25 \log_2 (0.25) + 0.25 \log_2 (0.25)]$

Since $\log_2 (0.5) = -1$ and $\log_2 (0.25) = -2$:

H = −[−0.5 − 0.5 − 0.5] = 1.5 bits.

$PP = 2^{1.5} \approx 2.828$.

Answer: Entropy = 1.5 bits; perplexity ≈ 2.828.

Part B (4 marks)

List the words that come after “the” and their probabilities. Find the entropy and perplexity of guessing the next word after “the”. What is the entropy of the next word after “sat”?

In the corpus, “the” appears 4 times. The words following “the” are:

• cat: 2 times, so P(cat | the) = 2/4 = 0.5.

• dog: 1 time, so P(dog | the) = 1/4 = 0.25.

• mat: 1 time, so P(mat | the) = 1/4 = 0.25.

Entropy after “the”:

$H = -[0.5 \log_2 (0.5) + 0.25 \log_2 (0.25) + 0.25 \log_2 (0.25)] = 1.5$ bits.

Perplexity after “the”: $PP = 2^{1.5} \approx 2.828$.

Entropy after “sat”: “sat” occurs twice and is followed by “on” both times, so P(on | sat) = 1.

$H = -1 \log_2 (1) = 0$ bits.

Final answers: after “the”: cat (0.5), dog (0.25), mat (0.25); entropy = 1.5 bits; perplexity ≈ 2.828; entropy after “sat” = 0 bits.